

Functional group	SMARTS string	From what
Ester	<chem>[#6][CX3](=O)[OX2H0][#6]</chem>	PET
Carboxylic acid	<chem>[CX3](=O)[OX2H]</chem>	TPA, N,N'-1,2-Ethanediybis[N-(carb oxymethyl)-glycine]
Alkane	<chem>[CX4;H3,H2]</chem>	N-Butyl-benzenesulfonamide, Tri-chloropropyl phosphate, Tris(2-butoxyethyl) phosphate, Triethyl phosphate, Tri- <i>n</i> -butyl phosphate, 2-Methyl-1,3-dioxolane, <i>n</i> -Nonanal, Toluene, <i>m/p</i> -Xylene, Limonene, Acetaldehyde, Ethylbenzene, <i>o</i> -Xylene, Dibutyl ester phosphoric acid, 1-Methyl-4-nitro-benzene, 2-Methyl-2-[(1-oxo-2-propen-1-yl)amino]-1-propanesulfonic acid, Bis(1-methyl-1-phenylethyl) peroxide
Alkene	<chem>[CX3]=[CX3]</chem>	Limonene, Styrene, 2-Methyl-2-[(1-oxo-2-propen-1-yl)amino]-1-propanesulfonic acid
Alcohol	<chem>[#6][OX2H]</chem>	Ethylene glycol, Diethylene glycol, Triethylene glycol
Arene	<chem>[cX3]1[cX3][cX3][cX3][cX3][cX3]1</chem>	PET,TPA, Benzophenone (diphenyl-methanone), N,N'-Diphenyl-guanidine, N-Butyl-benzenesulfonamide, Triphenyl phosphate, Benzaldehyde, Toluene, Benzene, <i>m/p</i> -Xylene, Styrene, Ethylbenzene, <i>o</i> -Xylene, 1-Methyl-4-nitro-benzene, Isoquinoline, 1 <i>H</i> -Benzotriazole, Bis(1-methyl-1-phenylethyl) peroxide
Amine	<chem>[NX3;H2,H1,H0;!\$(NC=O)]</chem>	Melamine (1,3,5-triazine-2,4,6-triamine), N,N'-Diphenyl-guanidine,

		N,N'-1,2-Ethanediyibis[N-(carb oxymethyl)-glycine]
Ketone	[#6][CX3](=O)[#6]	Benzophenone (diphenyl-methanone)
Ether	[OD2]([#6])[#6]	1,1'-Oxybis[2-methoxy-ethane] , Tris(2-butoxyethyl) phosphate, Diethylene glycol, 2-Methyl-1,3-dioxolane, Triethylene glycol
Imine	[\$([CX3]([#6])[#6]),\$([CX3H][#6]))=\$([NX2][#6]),\$([NX2H])]	Melamine (1,3,5-triazine-2,4,6-triamine), <i>N,N'</i> -Diphenyl-guanidine, Isoquinoline
Sulfonamide	[#16X4]([NX3])(=[OX1])(=[OX1])[#6]	<i>N</i> -Butyl-benzenesulfonamide
Acyl halide	[CX3](=[OX1])[F,Cl,Br,I]	Tri-chloropropyl phosphate (TCP, 3 isomers)
Phosphate	[\$([#16X4])(=[OX1])(=[OX1])([#6])[OX2H,OX1H0-]),\$([#16X4+2]([OX1-])([OX1-])([#6])[OX2H,OX1H0-])]	Triethyl phosphate, Tris(2-butoxyethyl) phosphate, Tri-chloropropyl phosphate, Triphenyl phosphate, Tri- <i>n</i> -butyl phosphate, Dibutyl ester phosphoric acid
Aldehyde	[CX3H1](=O)[#6,H]	Benzaldehyde, <i>n</i> -Nonanal, Formaldehyde, Acetaldehyde
Nitro group	[\$([NX3](=O)=O),\$([NX3+](=O)[O-])][!#8]	1-Methyl-4-nitro-benzene
Enamine	[NX3][CX3]=[CX3]	Isoquinoline
Azo compund	[#6][NX2]=[NX2][#6]	1 <i>H</i> -Benzotriazole
Sulfonic acid	[\$([#16X4])(=[OX1])(=[OX1])([#6])[OX2H,OX1H0-]),\$([#16X4+2]([OX1-])([OX1-])([#6])[OX2H,OX1H0-])]	2-Methyl-2-[(1-oxo-2-propen-1-yl)amino]-1-propanesulfonic acid
Amide	[NX3][CX3](=[OX1])[#6]	2-Methyl-2-[(1-oxo-2-propen-1-yl)amino]-1-propanesulfonic acid
Peroxide	[OX2,OX1-][OX2,OX1-]	Bis(1-methyl-1-phenylethyl) peroxide

Contaminants:

<https://pubs.rsc.org/en/content/articlelanding/2025/em/d5em00615e>

PMT plastic additives

Leachate screening <i>via</i> LC-QTOF-MS ^c	Melamine (1,3,5-triazine-2,4,6-triamine)	32%
	Benzophenone (diphenyl-methanone)	27%
	1,1'-Oxybis[2-methoxy-ethane]	23%
	<i>N,N</i> -Diphenyl-guanidine	23%
	<i>N</i> -Butyl-benzenesulfonamide	18%
	Dibutyl ester phosphoric acid	14%
	<i>N,N</i> -1,2-Ethanediyldis[<i>N</i> -(carboxymethyl)-glycine]	14%
	1-Methyl-4-nitro-benzene	9%
	Isoquinoline	9%
	1 <i>H</i> -Benzotriazole	5%
	2-Methyl-2-[(1-oxo-2-propen-1-yl)amino]-1-propanesulfonic acid	5%
	Bis(1-methyl-1-phenylethyl) peroxide	5%
OPEs		
Leachate screening <i>via</i> LC-QTOF-MS	Tri-chloropropyl phosphate (TCPP, 3 isomers) ^d	27%
	Tris(2-butoxyethyl) phosphate (TBEP) ^d	18%
	Triethyl phosphate (TEP) (LOQ = 0.122 mg kg ⁻¹)	14%
	Triphenyl phosphate (TPHP) (LOQ = 0.198 mg kg ⁻¹)	10%
	Tri- <i>n</i> -butyl phosphate (TNBP) (LOQ = 0.438 mg kg ⁻¹)	5%
NIAS		
Solvent extraction <i>via</i> GC-MS	Diethylene glycol	96%
	Ethylene glycol	69%
	2-Methyl-1,3-dioxolane	62%
	Benzaldehyde	46%

	<i>n</i> -Nonanal	38%
	Formaldehyde	38%
	Toluene	31%
	Benzene	31%
	<i>m/p</i> -Xylene	27%
	Limonene	19%
	Triethylene glycol	15%
	Styrene	15%
	Acetaldehyde	15%
	Ethylbenzene	4%
	<i>o</i> -Xylene	4%

Melamine is a precursor to resins

Tris(chloropropyl) phosphate (commonly abbreviated TCPP) is a chlorinated organophosphate flame retardant commonly added to polyurethane foams.

SMARTS codes:

https://www.daylight.com/dayhtml_tutorials/languages/smarts/smarts_examples.html